

Akhil S. Pillai

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RESEARCH INTERESTS

Shock wave interaction with structures and materials; compressible and two-phase flows; wave–structure coupling; high-fidelity, GPU-accelerated CFD on national HPC systems; where high-fidelity computation meets experiment in high-speed flows.

EDUCATION

PhD, Aerospace Engineering

2026 – present

Indian Institute of Technology Bombay

- Shock wave interaction with structures and materials.

B.Tech, Aerospace Engineering

2021 – 2025

Karunya Institute of Technology and Sciences, Coimbatore

- Thesis: *Numerical simulation of flow over a bluff body using a compressible multifluid six-equation formulation* — carried out at IIT Madras (host: Prof. A. Sameen); guide: Dr. Aldin Justin Sundaraj.

RESEARCH EXPERIENCE

Project Associate

Jul 2025 – Jul 2026

Department of Aerospace Engineering, IIT Madras — Prof. A. Sameen's group

- Six-equation two-fluid DNS of compressible gas–droplet flow past a circular cylinder ($Re = 100$, $Ma = 0.1$) with an in-house ~8,100-line Fortran 90 unstructured finite-volume solver (HLLC, eSSPRK(4,3), METIS/MPI, OpenACC GPU offload).
- Re-engineered the MPI halo exchange for GPUs — GPU-side pack/unpack into persistent flat buffers, one nonblocking `Isend/Irecv` pair per neighbour — for a ~11× end-to-end speedup with output bit-identical to the original code.
- Fixed a GPU data race in the residual reduction and a launch defect binding every rank to GPU 0 instead of one GPU per rank; corrected acoustically reflecting inlet/outlet conditions.
- Rebuilt a defective ICEM O-grid pipeline in GMSH; qualified a clean 3.5M-cell tetrahedral case (meshes to 28M cells evaluated). Ran under SLURM on A100 nodes and PARAM Shakti; post-processed in Tecplot 360 / PyTecplot and ParaView.

Research Intern

Dec 2024 – Jun 2025

Department of Aerospace Engineering, IIT Madras

- DNS of laminar vortex shedding and gas–droplet interface dynamics around a cylinder ($Re = 100, 150$; $Ma = 0.1$) with a six-equation two-fluid Fortran 90 code.
- Validated against the air–water shock-tube benchmark and Strouhal data; quantified droplet loading (Strouhal $0.221 \rightarrow 0.211$ at 0.1% dispersed volume fraction).

Summer Fellow

May – Jul 2024

IIT Madras Summer Fellowship Programme

- Machine learning for aerodynamic prediction: trained a CNN on XFOIL data to predict lift and drag from airfoil images, with ANN and PINN baselines in PyTorch (CUDA).

INDUSTRY EXPERIENCE

Internship Trainee

Jun 2023

Hindustan Aeronautics Limited, Bengaluru

- LCA Tejas production line: component fabrication, structural assembly, and system integration in a live defence-aviation environment.

Earlier internships

2021 – 2022

- OmSPACE Rocket & Exploration (launch vehicles, 2022); Brahmastra Aerospace Systems (industrial aerodynamics, 2022; aircraft vehicle design, 2021–22); XUANTA Developers (quantum computing, 2022); student delegate, 7th Bangalore Space Expo (CII, 2022).

TECHNICAL SKILLS

COMPUTING	Fortran 90, MPI, OpenACC, GPU computing, Python / PyTorch (CUDA), SLURM, LaTeX
SIMULATION	DNS, six-equation two-fluid modelling, Riemann solvers (HLLC), GMSH, ANSYS ICEM / Fluent
POST / CAD	Tecplot 360 / PyTecplot, ParaView, CATIA

LANGUAGES & MEMBERSHIP

English (professional working); Malayalam and Hindi (native/bilingual); Tamil (limited working). Student member, Aeronautical Society of India (2023–2026).